10 – Maximizing Probability of Choice

IEOR 145/245

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Course Project

- Details and sample projects on bCourses (see “Files”)

- Use the sheet for forming teams

Module 2: Uncertain Customer Choice

- We **assign/allocate** resources
  - Demand can be uncertain over time
  - Suitable for Ads and school bus problem

- In many settings, such as airlines and e-commerce, customers *choose* and we only decide a *set of options* to display
  - How do we model customer choice?
  - How do we show the right options?

What is a choice model?

- Choice model characterizes the probability that a product is chosen

$\phi(i, S) \coloneqq$ probability that a random customer chooses product $i \in S$, when shown assortment S

- $\phi(i, S) \in [0,1]$
- $\phi(S) \coloneqq \sum_{i \in S} \phi(i, S) =$ Probability that customer chooses something
- $\phi(S) + \phi(0, S) = 1$

RCM Example

- 2 products, $\Theta = \{1\ 2\ 0,\ 0\ 1\ 2,\ 1\ 0\ 2,\ 2\ 1\ 0,\ 0\ 2\ 1,\ 2\ 0\ 1\}$

- Define RCM through λ_σ: $\{0.5,\ \ 0,\ \ 0,\ \ 0,\ \ 0,\ \ 0.5\}$
  - i.e., $\lambda_{1\,2\,0} = \lambda_{2\,0\,1} = 0.5$ and all other rank orders have probability 0

- Given set $S = \{2\}$ i.e., only item 2 is available,

  customer chooses 2 with probability $\phi(2, \{2\}) = 1$
- Given $S = \{1\}$,

  $\phi(1, \{1\}) = 0.5$
- Given $S = \{1,2\}$,

  $\phi(1, \{1,2\}) = \phi(2, \{1,2\}) = 0.5$ and $\phi(\{1,2\}) = 1$

RUM Example

- 2 products
  - $u_1 = 1$ with probability (w.p.) 0.5
  - $u_1 = -1$ w.p. 0.5
  - $u_2 = 0.5$ w.p. 1

- Given set $S = \{2\}$ i.e., only item 2 is available,

$$\phi(2, \{2\}) = 1$$

- Given $S = \{1\}$,

$$\phi(1, \{1\}) = 0.5$$

- Given $S = \{1,2\}$,

$$\phi(1, \{1,2\}) = \phi(2, \{1,2\}) = 0.5 \text{ and } \phi(\{1,2\}) = 1$$

How do we use Choice Models?

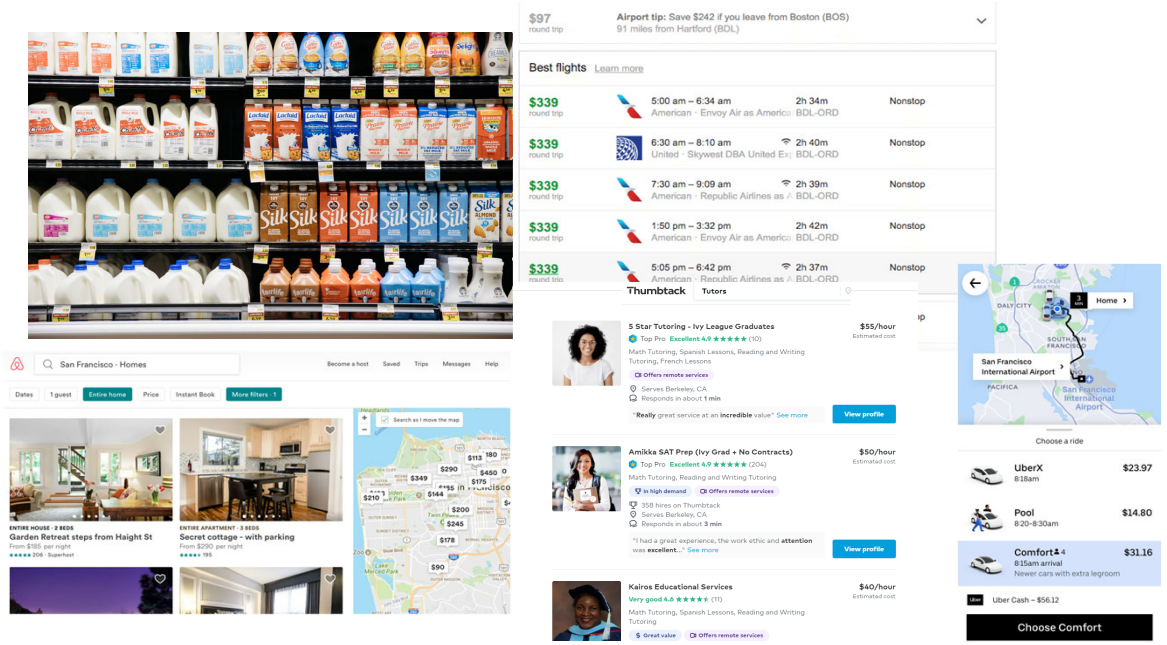
- When customers choose, we **engineer the options to offer**
  i.e., **optimize assortments**

- **Assumption:** we are given a choice model for the population

- We want to find the best assortment to show customers
  - What is best?

Best Assortment

- Thoughts?



Best Assortment: Two Objectives

1. Maximize probability that something is chosen

$$\max_{S \subseteq [n]} \phi(S)$$

On online platforms, this corresponds to *checkout rate or click rate*

Best Assortment: Two Objectives

1. Maximize probability that something is chosen

$$\max_{S \subseteq [n]} \phi(S)$$

2. Maximize expected revenue from choice

$$\max_{S \subseteq [n]} \sum_{i \in S} r_i \, \phi(i, S)$$

$(r_i)_{i \in [n]}$ is the vector of product prices (fixed till Module 3)

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$r_i = 1 \ \forall \, i$

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Two Objectives: Example

- Airline shows S={Basic Economy at \$100, Main Cabin at \$150}
  - $\phi(\text{Basic},S)=0.4,\ \phi(\text{Main},S)=0.2,\ \phi(0,S)=0.4$

- Objective 1, probability of a booking:

$$\phi(S)=0.4+0.2=0.6$$

- Objective 2, expected revenue: price × probability, added up

$$100\times0.4+150\times0.2=\$70$$

So we just maximize objective?

$$\max_{S \subseteq [n]} \phi(S) \quad \text{or} \quad \max_{S \subseteq [n]} \sum_{i \in S} r_i \, \phi(i, S)$$

- Suppose best was to include the entire universe of products
- Is this a viable solution?

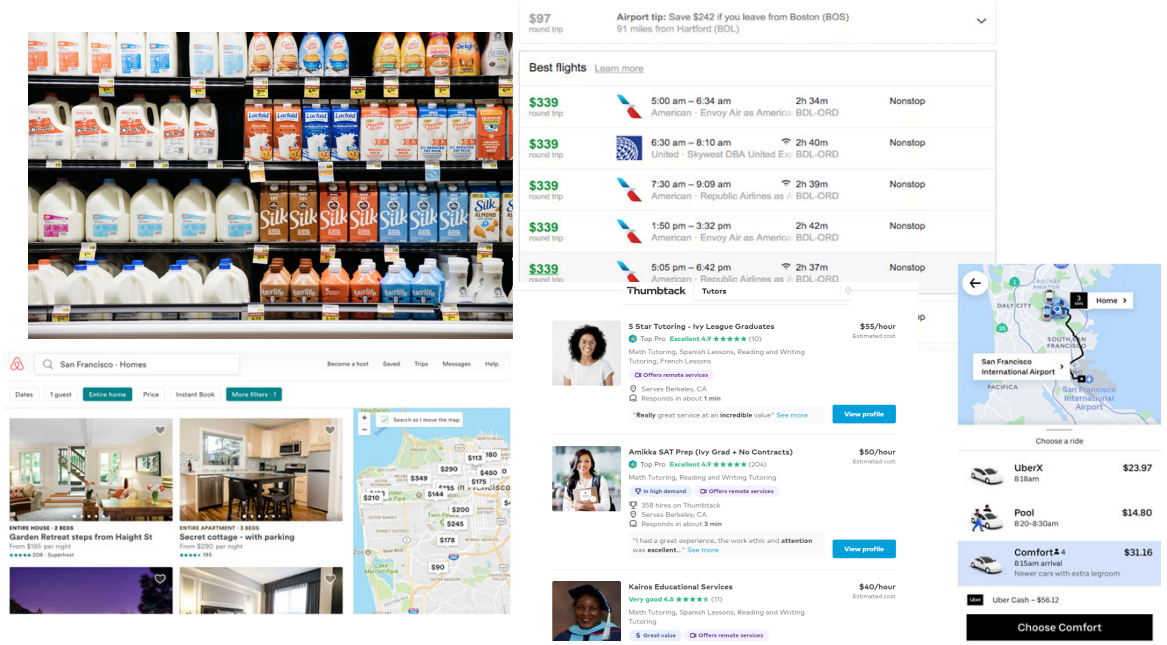
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- Suppose best was to include the entire universe of products
- Is this a viable solution?
- **Most practical situations have constraints**

Constraints

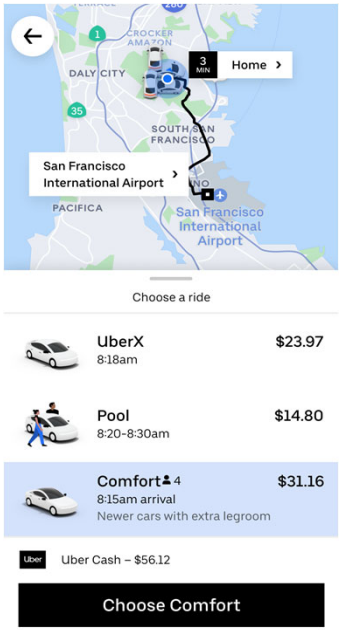
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Constraints

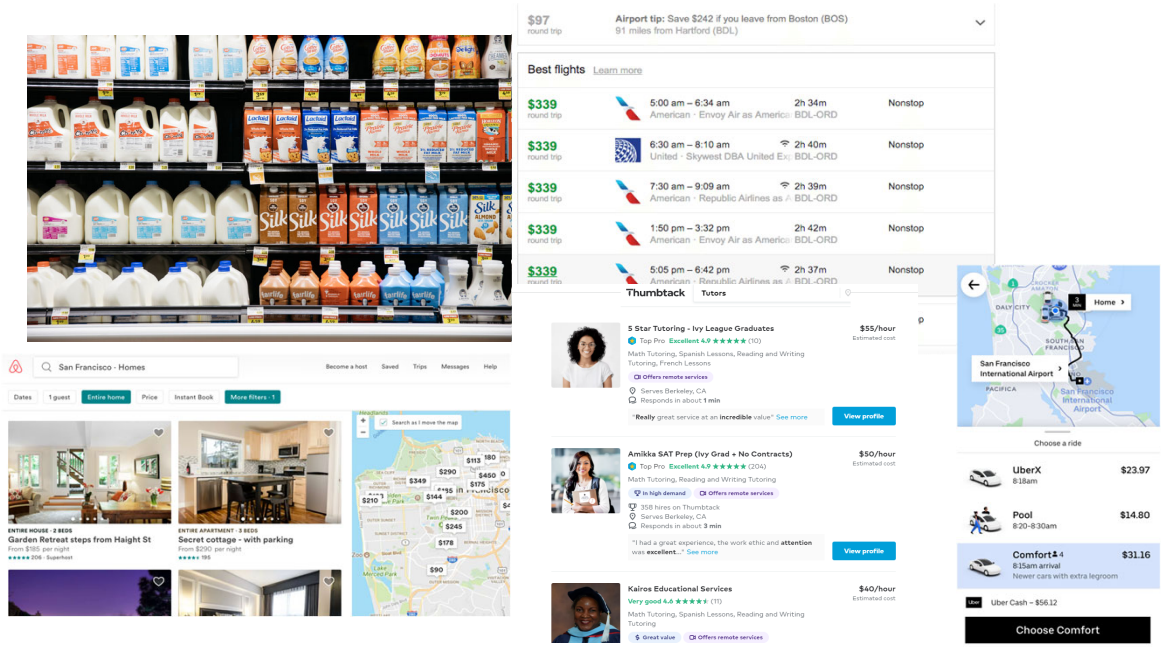
1. **Cardinality constraint:** Pick an assortment S of at most k products

$$|S| \leq k$$



$k = 3$

Cardinality good enough?



Constraints

1. **Cardinality constraint:** Pick an assortment of at most k products
2. **Capacity/Budget constraint:** Pick an assortment that takes at most B units of capacity/space

$$\sum_{i \in S} b_i \leq B$$

- $(b_i)_{i \in [n]}$ is the vector of space taken by individual items
- $b_i = 1 \ \forall \ i \in [n]$ captures cardinality constraint

e.g. Shelf with B=10 units of space
Each 1 Gallon needs 2 units of space, 0.5 needs 1
$S_1 = \{3$ x 1 Gallon, 4 x half gallon$\}$ is **feasible**
$S_1 = \{6$ x 1 Gallon$\}$ is **infeasible**



Constrained Assortment Optimization: Two Objectives

1. Maximize probability that something is chosen

$$\max_{S \subseteq [n] \mid S \in \mathcal{F}} \phi(S)$$

2. Maximize expected revenue from choice

$$\max_{S \subseteq [n] \mid S \in \mathcal{F}} \sum_{i \in S} r_i \, \phi(i, S)$$

$r_i = 1 \ \forall \, i$

$[n]=\{1,2,\ldots,n\}$ is a large universe (catalogue) of products
$S \subseteq [n]$: S is any subset, i.e., any assortment we could show
$S \in \mathcal{F}$: S is feasible, i.e., obeys the constraint (e.g., fits on the shelf)

Maximizing Probability of Selection

$(r_i = 1 \ \forall i \in [n])$

Cardinality Constrained Optimization

Find assortment such that

$$\max_{S \subseteq [n] \mid |\boldsymbol{S}| \leq \boldsymbol{k}} \phi(S)$$

- Optimal assortment?
  - Suppose $n = 10$, can evaluate 2^{10} candidates easily
  - **Larger $\boldsymbol{n}$**? (in reality, n between hundreds to millions)
  - In general, a hard problem

How Many Assortments?

- Each product is either in or out → 2^n possible assortments
  - $n=10$: $2^{10}=1{,}024$ → easy to check them all
  - $n=100$: $2^{100}\approx 1.3\times 10^{30}$
  - $n=1000$: about 10^{301}

IP – 1 for Cardinality Constraint

- $y_S \in \{0,1\}$ decides if assortment S is chosen

$$\max \sum_{S \mid |S| \leq k} \phi(S)\, y_S$$

$$s.t. \sum_{S \mid |S| \leq k} y_S \leq 1$$

IP – 1 for Cardinality Constraint [Intractable]

- $y_S \in \{0,1\}$ decides if assortment S is chosen

$$\max \sum_{S \mid |S| \leq k} \phi(S)\, y_S$$

$$s.t. \sum_{S \mid |S| \leq k} y_S \leq 1$$

$O(n^k)$ decision variables!

IP – 2 for Cardinality Constraint

- $x_i \in \{0,1\}$ decides if product i is chosen
- $S_x = \{i \mid i \in [n],\ x_i = 1\}$

$$\max_{(x_i)_{i\in[n]}} \phi(S_x)$$
$$s.t. \quad \sum_i x_i \leq k$$

IP – 2 for Cardinality Constraint

- $x_i \in \{0,1\}$ decides if product i is chosen
- $S_x = \{i \mid i \in [n],\ x_i = 1\}$

$$\max_{(x_i)_{i\in[n]}} \phi(S_x) = \sum_{i[n]} \phi(i, S_x)\ x_i$$

$$s.t. \quad \sum_i x_i \leq k$$

Non-linear objective
e.g. $(1 - x_j)x_i$

IP – 2 Objective Example

- Recall our RCM example

  (half population has ranking 1 2 0, other half 2 0 1):

  - 1 2 0 customers buy 1 if $x_1=1$ and product 2 if $x_1=0$ and $x_2=1$
  - 2 0 1 customers buy 2 if $x_2=1$

$$\phi(S_x)=0.5[x_1+(1-x_1)x_2]+0.5x_2$$

Heuristics for Probability Maximization with Cardinality Constraint

Heuristic – I

$$\max_{S \subseteq [n] \mid |\boldsymbol{S}| \leq \boldsymbol{k}} \sum_{i \in S} \phi(i, S)$$

- Let's use the inequality: $\phi(i, S) \leq \phi(i, \{i\})$
  - Follows from the substitutability property (last lecture) (how?)

- Can we solve: $\max_{S \subseteq [n] \mid |\boldsymbol{S}| \leq \boldsymbol{k}} \sum_{i \in S} \phi(i, \{i\})$?

$\phi(i,\{i\})$ = how popular product i is when shown **alone**

Heuristic – I

$$\max_{S \subseteq [n] \mid |S| \leq k} \sum_{i \in S} \phi(i, S)$$

- $\sum_{i \in S} \phi(i, S) \leq \sum_{i \in S} \phi(i, \{i\})$
  - From substitutability of ϕ (how?)
- Can we solve $\max_{S \subseteq [n] \mid |S| \leq k} \sum_{i \in S} \phi(i, \{i\})$?
- Yes, pick k products with the highest values $\{\phi(i, \{i\})\}_{i \in [n]}$

Heuristic – I in Words

- Heuristic I: rank products by solo popularity, show the top k
  - Like a bookstore putting just all Harry Potter novels on the front table
  - Recall that $\phi(i,\{i\})$ = how popular product i is when shown **alone**

- Replaces the real objective $\sum_{i \in S} \phi(i,S)$ by the simpler $\sum_{i \in S} \phi(i,\{i\})$
  - Simple: ignores substitution (competition) between products

Is this a good algorithm?

Solve $\max_{S \subseteq [n] \mid |\boldsymbol{S}| \leq \boldsymbol{k}} \sum_{i \in S} \phi(i, S)$

- Algorithm: pick k products with highest value of $\phi(i, \{i\})$

Is this a good algorithm? - NO

Solve $\max_{S \subseteq [n] \mid |\boldsymbol{S}| \leq \boldsymbol{k}} \sum_{i \in S} \phi(i, S)$

- Algorithm: pick k products with highest value of $\phi(i, \{i\})$
- E.g.: $\lambda_{1,\,2,\,3,\,4,..10,0} = 0.1,\ \lambda_{11,\,0} = \lambda_{12,\,0} = \lambda_{13,\,0} = \cdots = \lambda_{19,\,0} = 0.09$
- For $k = 10$, algorithm picks $\{1,2,3, \ldots 10\}$
  - $\phi(\{1,2, \ldots 10\}) = 0.1$
- Optimal: $\phi(\{1,11,12,13, \ldots 19\}) = 0.91$

Why Heuristic – I Fails

- Products 1–10: ten nearly identical fiction novels
  - The same 10% of customers (list 1 2 3 … 10 0) want them
- Products 11–19: nine different genres, each with its own 9% of fans
- Heuristic shows 10 fiction novels
  - Reaches the same 10% of population
- Better: 1 fiction + 9 other genres: $0.1+9\times0.09=0.91$

- Select products that attract the **different** customers

Heuristic – II: Greedy Algorithm

$$\max_{S \subseteq [n] \mid |\boldsymbol{S}| \leq \boldsymbol{k}} \phi(S)$$

- Idea: Iteratively pick best *accounting for previous iterations*

- Define marginal value function,

$$\Delta\phi(e|S) = \phi(S + \{e\}) - \phi(S)$$

captures increase in
probability on adding e

Heuristic – II: Greedy Algorithm

$$\max_{S \subseteq [n] \mid |\boldsymbol{S}| \leq \boldsymbol{k}} \phi(S)$$

- Idea: Iteratively pick best *accounting for previous iterations*
- Define marginal value function,

$$\Delta\phi(e|S) = \phi(S + \{e\}) - \phi(S)$$

1. **Build solution in $\boldsymbol{k}$ steps**
2. **Always add element with maximum marginal value to current set**

$$\text{Add } \operatorname{argmax}_{e \in [n] \setminus S} \Delta\phi(e|S)$$

Example

- E.g.:$\lambda_{1,\,2,\,3,\,\,4,..10,0} = 0.1,\ \lambda_{11,\,0} = \lambda_{12,\,0} = \lambda_{13,\,0} = \cdots = \lambda_{19,\,0} = 0.09$

- For $k = 10,$ what does Greedy pick?

Example

- E.g.:$\lambda_{1,\,2,\,3,\,\,4,..10,0} = 0.1,\ \lambda_{11,\,0} = \lambda_{12,\,0} = \lambda_{13,\,0} = \cdots = \lambda_{19,\,0} = 0.09$

- For $k = 10,$ what does Greedy pick?
- **Step 1:** product 1
- **Step 2:** product 11
- **Step 3:** product 12 and so on…

- Greedy picks the optimal solution $\{1,11,12, \ldots 19\}$

Example: Greedy Step by Step

- $\Delta\phi(e|S)$ = extra customers won by adding e to what is already shown
- **Step 1** ($S=\emptyset$): products 1–10 add 0.1, products 11–19 add 0.09

  → add 1

- **Step 2** ($S=\{1\}$): products 2–10 now add 0 (their fans already buy 1); 11 adds 0.09

  → add 11

- **Steps 3–10**: add 12, 13, …, 19, each add 0.09
- Total: $0.1+9\times0.09=0.91$, the optimal value

Is Greedy Always Good?

$$\Delta\phi(e|S) = \phi(S + \{e\}) - \phi(S)$$

1. Build solution in k steps
2. Always add element with maximum marginal value to current set

- Does it always get the optimal solution?

Is Greedy Always Good?

$$\Delta\phi(e|S) = \phi(S + \{e\}) - \phi(S)$$

1. Build solution in k steps
2. Always add element with maximum marginal value to current set

- Does it always get the optimal solution?
  - Typically, no (the problem is hard to solve quickly)
  - Is there an example of suboptimal situation?
  - How do we know if Greedy is a “good” algorithm?